

WOO JAE KIM

(+82) 010-4584-3490 ◇ woojae1123@gmail.com ◇ wkim97@kaist.ac.kr
wkim97.github.io ◇ github.com/wkim97 ◇ [Google Scholar](#)

Trustworthy AI · Adversarial Robustness · Visual & 3D Content Protection · Embodied AI
School of Computing, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea

EDUCATION

Ph.D., School of Computing Mar. 2023 – Feb. 2027 (Expected)
KAIST, Daejeon, Korea *Advisor: Sung-Eui Yoon*
Dissertation (working title): Robust and Reliable AI under Non-Ideal Conditions
Committee: Sung-Eui Yoon (chair), Jinwoo Shin, Sooel Son, Seunghoon Hong, Hwanjun Song

M.S., School of Computing Mar. 2021 – Feb. 2023
KAIST, Daejeon, Korea *Advisor: Sung-Eui Yoon*

B.S., School of Computing Sep. 2016 – Feb. 2021
KAIST, Daejeon, Korea *Minor in Electrical Engineering*

High School Diploma Sep. 2012 – May 2016
Northview High School, Johns Creek, GA, USA

EXPERIENCE

Research Intern, NAVER LABS Apr. 2026 – Oct. 2026
Seongnam, Korea *Robotics and Embodied AI*

- Building visually grounded refinement strategies for vision-language-navigation (VLN) under imperfect user instructions.

RESEARCH INTERESTS

Core focus: Building **robust and reliable AI under non-ideal conditions**. I diagnose how models fail in real deployment and design targeted mechanisms that restore trustworthy behavior, carrying this principle across **AI's evolving paradigms**:

- **2D vision** (2021–) — **adversarial robustness:** expose vulnerabilities via transferable attacks and harden models against single- and multiple-threat perturbations (ICIP'22 Oral, CVPR'23 Highlights, ECCV'26).
- **3D vision** (2024–) — **content protection:** repurpose adversarial perturbations to shield neural radiance fields and 3D Gaussian splatting from unauthorized use (BMVC'25).
- **Generative & language models** (2025–) — **controllable, transparent generation:** align large language models for controllable outputs that transparently disclose advertising at zero additional cost (NeurIPS'26 under review).
- **Embodied AI** (2026–) — **reliable embodied agents:** visually ground vision-language-navigation to recover user intent from ambiguous, imperfect instructions (In progress).

PUBLICATIONS

(Underlined denotes the author; * denotes equal contribution.)

21. Visually Grounded Instruction Refinement for Vision-Language-Navigation under Imperfect Instructions
Woo Jae Kim et al.
In progress

- 20. AdKnob: Ad Intensity Control and Labeling for LLM-Native Advertising**
Woo Jae Kim, Seongho Keum, Joonsung Jeon, Suhyeon Ha, Sooel Son, and Sung-Eui Yoon.
NeurIPS, 2026 (under review)
- 19. AHEAD: Attention HEads for Adversarial Detection and Defense in GUI Grounding**
Geumyoung Jung, Suhyeon Ha, Woo Jae Kim, and Sung-Eui Yoon.
NeurIPS, 2026 (under review)
- 18. EvBS: Event-guided Blur Synthesis for Domain-adaptive Motion Deblurring**
Junsik Jung, Seokryun Choi, Yoonki Cho, Woo Jae Kim, Andrew Jeong, and Sung-Eui Yoon.
ACM MM, 2026 (under review)
- 17. RoME: Robust Mixture of Low-Rank Experts against Multiple Adversarial Perturbations**
Woo Jae Kim, Kyle Min, Suhyeon Ha, Joonsung Jeon, and Sung-Eui Yoon.
ECCV, 2026
- 16. APT: Anchor-aligned Perturbations for Tamper Localization in Fully Regenerated Images**
Suhyeon Ha, Woo Jae Kim, Joonsung Jeon, Sooel Son, and Sung-Eui Yoon.
ECCV, 2026
- 15. Beyond the Patch: Exploring Vulnerabilities of Visuomotor Policies via Viewpoint-Consistent 3D Adversarial Object**
Chanmi Lee, Minsung Yoon, Woo Jae Kim, Sebin Lee, and Sung-Eui Yoon.
ICRA, 2026
- 14. Radiometrically Consistent Gaussian Surfels for Inverse Rendering**
Kyu Beom Han, Jaeyoon Kim, Woo Jae Kim, Jinhwan Seo, and Sung-Eui Yoon.
ICLR, 2026 **Oral paper**
- 13. No Caption, No Problem: Caption-Free Membership Inference via Model-Fitted Embeddings**
Joonsung Jeon, Woo Jae Kim, Suhyeon Ha, Sooel Son, and Sung-Eui Yoon.
ICLR, 2026
- 12. AegisRF: Adversarial Perturbations Guided with Sensitivity for Protecting Intellectual Property of Neural Radiance Fields**
Woo Jae Kim, Kyu Beom Han, Yoonki Cho, Youngju Na, Junsik Jung, Sooel Son, and Sung-Eui Yoon.
BMVC, 2025
Also accepted to ECCV Workshops 2024 (Computer Vision for Metaverse Workshop)
- 11. Learning Event-guided Exposure-agnostic Video Frame Interpolation via Adaptive Feature Blending**
Junsik Jung, Yoonki Cho, Woo Jae Kim, Lin Wang, and Sung-Eui Yoon.
BMVC, 2025
- 10. Pose-free 3D Gaussian Splatting via Shape-Ray Estimation**
Youngju Na*, Taeyeon Kim*, Jumin Lee, Kyu Beom Han, Woo Jae Kim, and Sung-Eui Yoon.
ICIP, 2025 **Best Student Paper Award** (Top 0.16%)
- 9. AdvPaint: Protecting Images from Inpainting Manipulation via Adversarial Attention Disruption**
Joonsung Jeon, Woo Jae Kim, Suhyeon Ha, Sooel Son, and Sung-Eui Yoon.
ICLR, 2025
- 8. Generalizable Person Re-identification via Balancing Alignment and Uniformity**
Yoonki Cho, Jaeyoon Kim, Woo Jae Kim, Junsik Jung, and Sung-Eui Yoon.
NeurIPS, 2024

7. UFORecon: Generalizable Sparse-View Surface Reconstruction from Arbitrary and Unfavorable Sets
Youngju Na, **Woo Jae Kim**, Kyu Beom Han, Suhyeon Ha, and Sung-Eui Yoon.

CVPR, 2024

6. Towards Content-based Pixel Retrieval in Revisited Oxford and Paris

Guoyuan An, **Woo Jae Kim**, Saelyne Yang, Rong Li, Yuchi Huo, and Sung-Eui Yoon.

ICCV, 2023

5. Feature Separation and Recalibration for Adversarial Robustness

Woo Jae Kim, Yoonki Cho, Junsik Jung, and Sung-Eui Yoon.

CVPR, 2023 **Highlights paper** (~2.5% acceptance rate)

4. Pixel-wise Guidance for Utilizing Auxiliary Features in Monte Carlo Denoising

Kyu Beom Han, Olivia G. Odenthal, **Woo Jae Kim**, and Sung-Eui Yoon.

i3D, 2023

3. Diverse Generative Perturbations on Attention Space for Transferable Adversarial Attacks

Woo Jae Kim, Seunghoon Hong, and Sung-Eui Yoon.

ICIP, 2022 **Oral paper** (~10% acceptance rate)

2. Part-based Pseudo Label Refinement for Unsupervised Person Re-identification

Yoonki Cho, **Woo Jae Kim**, Seunghoon Hong, and Sung-Eui Yoon.

CVPR, 2022

1. Deep Video Inpainting Guided by Audio-Visual Self-Supervision

Kyuyeon Kim, Junsik Jung*, **Woo Jae Kim***, and Sung-Eui Yoon.

ICASSP, 2022

PATENTS

3. Method and System for Separating and Recalibrating Feature for Adversarial Robustness

Woo Jae Kim, Yoonki Cho, Junsik Jung, and Sung-Eui Yoon.

US Patent Pending (Application No. 18/639,290), Filed Feb. 2024

Japan Patent Granted (Patent No. 7628350), Jan. 2025

Korea Patent Pending (Application No. 10-2024-0002170), Filed Jan. 2024

Selected as a **strategic technology** by the KAIST Technology Licensing Center

2. Method and Computer Device for Generalizable Sparse-View Surface Reconstruction from Arbitrary and Unfavorable Sets

Youngju Na, **Woo Jae Kim**, Kyu Beom Han, Suhyeon Ha, and Sung-Eui Yoon.

Korea Patent Pending (Application No. 10-2024-0071859), Filed May 2024

1. Pixel-wise Guidance for Utilizing Auxiliary Features in Monte Carlo Denoising

Kyu Beom Han, **Woo Jae Kim**, and Sung-Eui Yoon.

Korea Patent Pending (Application No. 10-2023-0190779), Filed Dec. 2023

FELLOWSHIPS & AWARDS

Sep. 2025 **Best Student Paper Award**, International Conference on Image Processing (ICIP) 2025 (as co-author).

Nov. 2023 **Recipient**, Qualcomm Innovation Fellowship.

Feb. 2022 **Paper Award**, 34th Workshop on Image Processing and Image Understanding (IPIU).

Aug. 2023, Feb. 2022 **Best TA Award**, School of Computing, KAIST.

Feb. 2021 **Grand Prix Award (1st place)**, Undergraduate Research Program (URP), KAIST.

INVITED TALKS & PRESENTATIONS

Jan. 2024	Invited talk on adversarial robustness (FSR) at Qualcomm Korea.
Feb. 2023	Oral and poster presentations on adversarial robustness (FSR), Korean Conference on Computer Vision (KCCV) 2023.

PROFESSIONAL SERVICE

Conference Reviewer: CVPR (2023–2026) · ICCV (2023–2025) · ECCV (2024–2026) · NeurIPS (2025–2026) · ACM MM (2025) · WACV (2025) · ACCV (2024)